Text Summarizer using Machine Learning

ABSTRACT

In this new era, where tremendous information is available on the internet, it is most important to provide the improved mechanism to extract the information quickly and most efficiently. It is very difficult for human beings to manually extract the summary of a large documents of text. There are plenty of material available on the internet. So, there is a problem of searching for relevant documents from the number of documents available, and absorbing relevant information from it. In order to solve the above two problems, the automatic text summarization is very much necessary. As the project title suggests, Text Summarizer is a web based application which helps in summarizing the text i.e., compressing them into a shorter version preserving its overall meanings. We can upload our data and this application gives us the summary of that data in as many numbers of lines as we want. This project helps us to give a brief context of the large content and identify the interest area. This is a Python project. The branch of NLP that task that extracts the relevant topic from the textual document is the automatic text summarizer. It does the task of converting the long textual document into short fluent summaries. There are generally two ways of summarizing text using automatic text summarizer, first is using extractive text summarizer and another is abstractive text summarizer. In this project not only a text summarization is done, it also includes pdf summarization and URL summarization. We can extract main keywords in a large text, also possible collect reference website and better understanding with image. There are many datasets available for summarization. The datasets domain ranges from News articles, Reddit and Wikipedia

Existing System

Text summarization has traditionally been focused on text input. Various methods for summarization were proposed which include extraction based, abstraction-based, maximum entropy-based and aided summarization. These methods use linguistic and natural language processing techniques. The steps followed are interpretation of the source text to obtain a text representation, transformation of the text representation into a summary representation, and finally, generation of the summary text from the summary representation. Most of the existing methods for document summarization use the information present in the given document. Comments left by readers on Web documents contain valuable information that can be utilized in various information retrieval tasks including document search, visualization and summarization. In this project, we study the problem of comments-oriented blog summarization and aim to summarize a blog post by considering not only its content, but also the comments left by its readers. In this project, we aim to extract representative sentences from a blog post that best represent the topics discussed among its comments. Drawbacks in Existing System: - (I) Summary is less accurate (II) Time constraint is less (III) Computational speed is more. (IV) It uses Abstractive Summarization.

Proposed System

Reading the large content of text available online is challenging for the users as it consumes a large amount of time. The proposed system, Automatic Text Summarization is much more practical and applicable in real time. The input text is processed using natural language processing and processed input is converted into vector form using word embedding. Word embedding is the collective name for a set of language modelling and feature learning techniques in NLP where words or phrases from the vocabulary are mapped to vectors of real numbers. Sentence ranking is done between sentences to extract higher ranked sentence, which forms the extractive summary of the input. The summarized text is then analysed using polarity and subjectivity parameters. The summarized text is also subjected to speech conversion. Text Summarizer using Machine Learning 5 Advantages of Proposed System: - (i) Our Algorithm executes with good performance because we are executing in a distributed environment. (ii) Time Constraint is less (iii)Computational Speed is more. (iv) Accurate Summary (v) It uses Extractive Summarization.

H/W SYSTEM REQUIREMENT

- Processors :Intel core I3 or above
- Keyboard :104 keys
- Display :15" Color Monitor
- RAM : 4Gb or Above
- Storage : 512GB and above

S/W SYSTEM REQUIREMENTS

- Operating System :Windows 10 and above
- Front End :HTML,CSS,JAVASCRIPT
- Backend :Mysql,Python,PHP
- IDE :Notepad++
- Software Kit :XAMPP

MODULES

1. User Modules

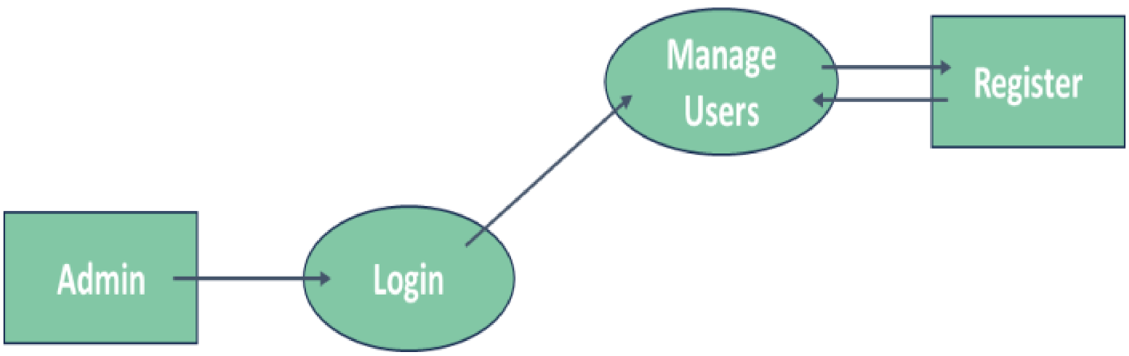
- Register
- Login
- Upload PDF
- Enter URL
- View Result
- Logout

DIAGRAMS

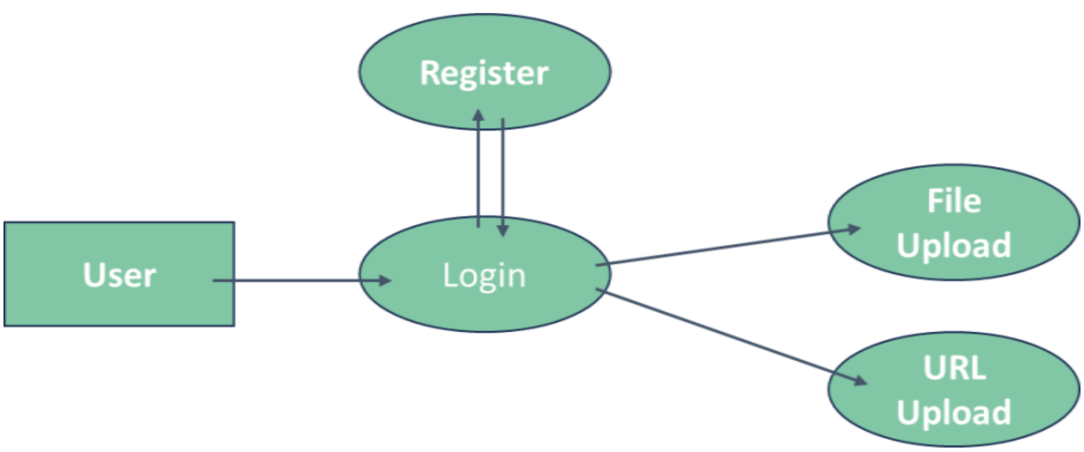
DATA FLOW DIAGRAM

Level 0

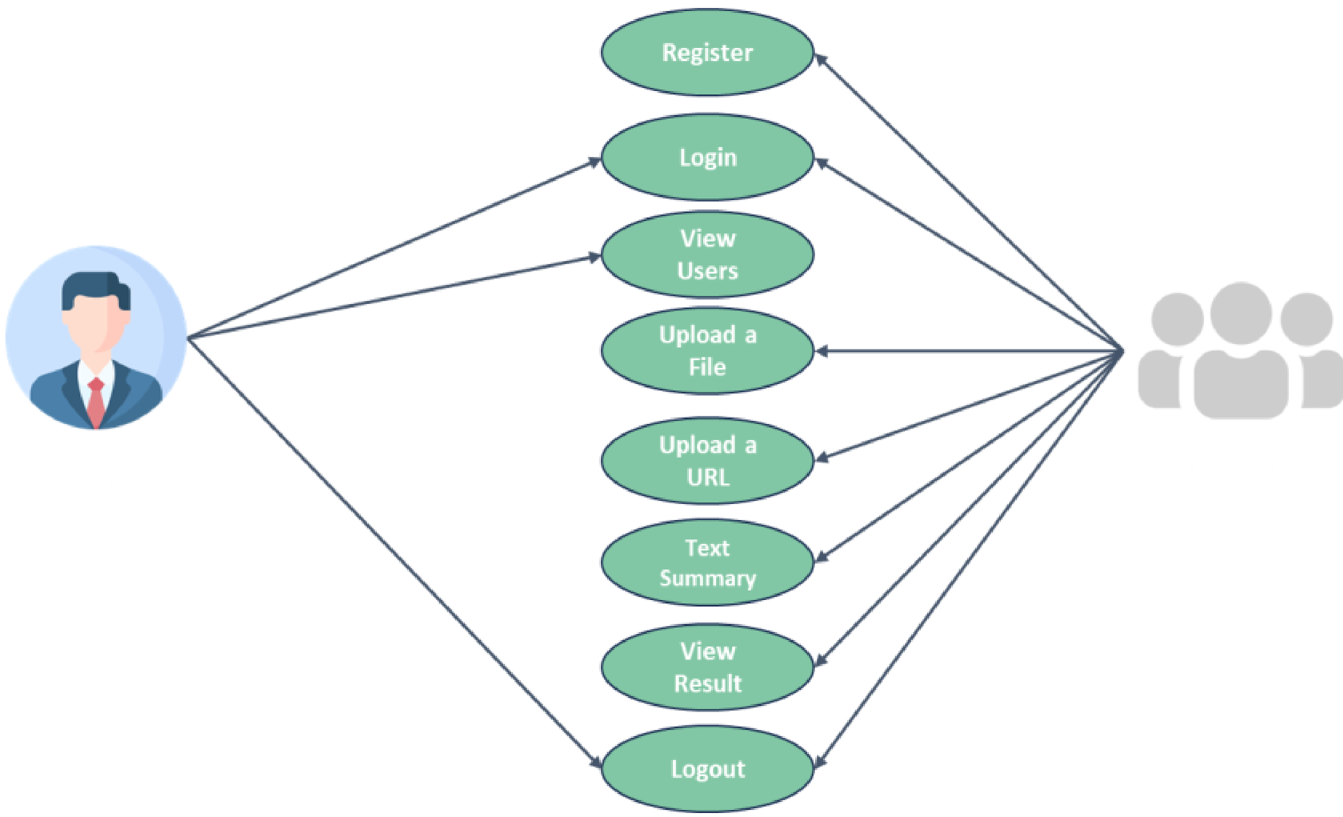
Level 1



Level 2

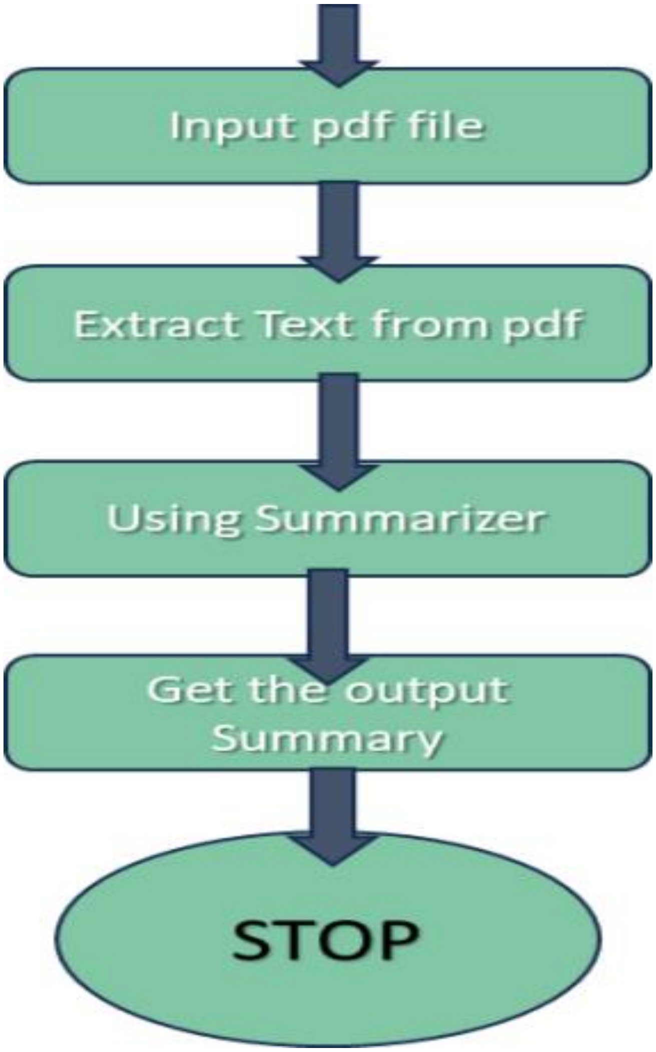


USECASE DIAGRAM



FLOW CHART

1) PDF to summary



2) Extract summary from url